

Turtle Movement & Animation

Experimenting with movement

- Do we understand what the `click_handler` in `experiment.py` does?
- If not, we can set the speed to 1 (slowest) to watch it
- Let's add a 180 degree turn at the end and see what changes
- Now add a loop that runs twice, so we return to our initial state
- If we change the turn to 90 degrees, how many repeats does the loop need? What about $(360 / 7)$ degrees?
- This is starting to get large, so we should separate it into its own function

Using multiple parameters

- We can define some parameters for this new function
 - Number of sides - how many times to repeat the loop, and the denominator of the rotation
 - Size of the shape - how much to move forward
 - Size of the backup - how much to move backward

For loops to make animations

- You can use a for loop's iterating variable as an argument to this function to create an animation
- To make the animation smoother, you can stop the window from updating except when you want it to
 - Call `window.tracer(0)` to stop the window from updating
 - Call `window.update()` to manually update the window

Other ways to use for loops

- What if we plug a for loop's iterating variable into forward movement?
- Can we change these numbers to make a spiral?

Using two turtles

- The biggest impediment to using two turtles is that the onclick() function doesn't specify which turtle was clicked on
- So, you either need to create different event_handler functions for each turtle (which is tedious if you want turtles to behave similar) or you need one event_handler to handle all turtles at the same time
- If each turtle has its own event_handler, then each event_handler runs completely before allowing the other turtle to act
- Another option is to split individual frames of the animation by using window.ontimer()